

Anatomy of a False Edge

A Falsification of a Microstructure Market-Making Strategy
on NSE Equity Futures

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Abstract

This report documents a self-directed research project that built a live paper-trading system for a microstructure market-making strategy on National Stock Exchange (NSE) equity futures, and then falsified it. Under the simulator's optimistic touch-fill model the strategy appeared strongly profitable. A markout and mid-to-mid decomposition, computed directly from the stored order book, shows that the entire reported profit was an artifact of assumed bid-ask spread capture: priced at the mid, the strategy lost Rs 10.14 lakh and its win rate collapsed from 77% to 12.6%, exposing a *negative* directional edge. Break-even requires retaining at least 81% of the theoretical spread, a level a non-colocated retail participant cannot reach given back-of-queue priority, roughly 15 ms latency, and a securities transaction tax that dominates round-trip cost. Overfitting-aware evaluation (Deflated Sharpe Ratio, Probability of Backtest Overfitting) promoted no variant. The project was halted at this documented verdict and its design pivoted toward taker-side, model-based strategies at horizons where latency does not decide the outcome. No real capital was deployed.

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1 Introduction and motivation

This is a falsification study. It documents how a systematic trading strategy that looks profitable in simulation can be an artifact of its fill model, how to detect that before committing capital, and why the resulting verdict was specific to the trading seat rather than to the idea. The deliverable is the method and the decision, not a strategy that works.

The strategy is a passive market maker driven by a single signal, the microprice deviation from the level-one order book,

$$\text{microprice} = \frac{\text{bid}_{\text{px}} \cdot \text{ask}_{\text{qty}} + \text{ask}_{\text{px}} \cdot \text{bid}_{\text{qty}}}{\text{bid}_{\text{qty}} + \text{ask}_{\text{qty}}}, \quad \text{deviation} = \text{microprice} - \text{mid},$$

an imbalance-weighted fair value that leans toward the heavier side of the book [1]. The strategy posts a passive limit order at the bid or ask when the deviation exceeds a floor and an economic edge gate is satisfied (the captured half-spread must cover the per-share round-trip fee), then exits passively at the opposite side, with taker fallbacks on a hard stop and a hold-time limit. The intent is to earn the spread rather than pay it.

2 Data and experimental design

A collector on a Mumbai cloud VPS archives the full NSE 20-level tick-by-tick order book for a liquid futures universe, stored as date-partitioned Parquet. This order book is the raw material for every result below.

Strategy variants were compared *live and head-to-head* rather than by backtest, for a specific reason: a compacted-data replay of the strategy missed a live trading day by Rs 11k, which localised the unreliability to the fill model, the one component a backtest cannot get right. The system therefore ran a multi-arm harness: seven “arms” (a control plus challengers varying stop width, universe, selectivity, and exit logic) fed by one shared depth connection, each with isolated brokers, an independent risk governor, and namespaced logs, so arms never contaminate one another. Champion-versus-challenger on identical live data is the only trustworthy way to compare variants when the simulator itself is suspect. The analysis below covers twelve clean trading days.

3 The apparent result, and the first red flag

On the simulator’s touch-fill model (a passive order fills the instant price touches its level) the strategy looked strongly profitable, with a win rate near 77%. The first sign that this was not what it seemed was a unit check: the strategy captured roughly 4.5 basis points of gross profit per trade, while the half-spread on these instruments is only about 0.2 to 0.4 basis points. A market maker’s edge *is* the spread; capturing fifteen to twenty-five times the spread means the profit is coming from *directional* price moves, not from providing liquidity, and directional capture is precisely where an optimistic fill model is least trustworthy.

4 Method: the mid-to-mid decomposition

Two quantities, both computed straight from the stored order book and therefore independent of the fill model, separate assumed spread capture from realised price movement.

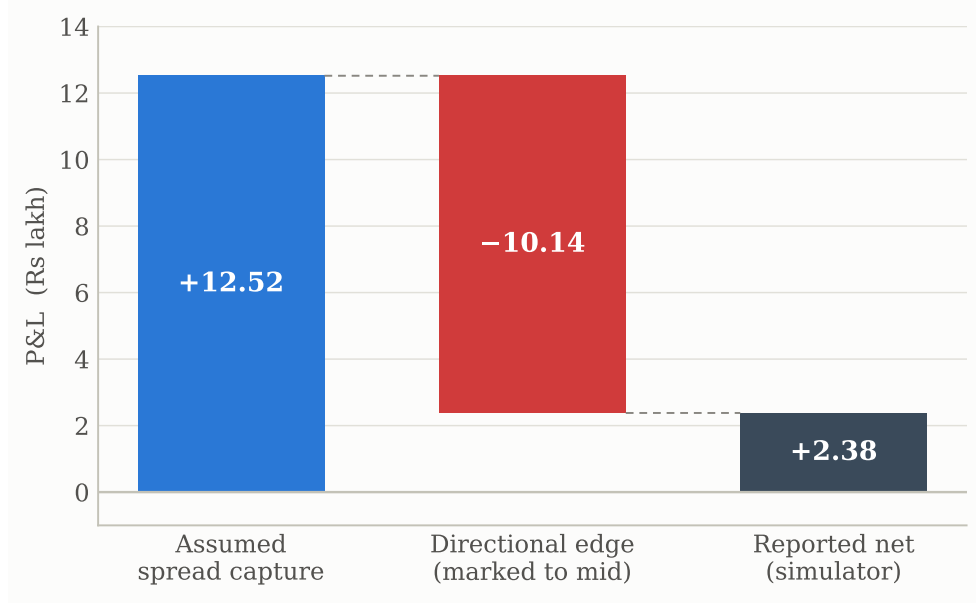


Figure 1. The simulated profit decomposed. Marking both legs to the mid removes the assumed bid-ask capture and reveals a large negative directional result. The reported net is a thin, fragile residual on top of a spread assumption five times its size.

Markout. For a fill at price p in direction $d \in \{+1, -1\}$,

$$\text{markout}(h) = d (\text{mid}(t_{\text{fill}} + h) - p).$$

At $h = 0$ this is the captured half-spread (a passive fill sits inside the mid); as h grows it reveals any adverse drift after the fill.

Mid-to-mid profit and loss. Re-pricing *both* legs of a round trip at the mid,

$$\text{P\&L}_{\text{mid}} = d (\text{mid}_{\text{exit}} - \text{mid}_{\text{entry}}) \cdot \text{qty} - \text{fee},$$

removes all spread capture and leaves the pure directional result. The two views satisfy the identity

$$\text{P\&L}_{\text{sim}} = \text{P\&L}_{\text{mid}} + (\text{assumed spread capture}),$$

so decomposing the simulated profit isolates exactly how much of it was the spread the model handed itself for free.

5 Results

The decomposition is unambiguous (Figure 1). Of the simulated +Rs 2.38 lakh, *all* of it, and Rs 10.14 lakh more, is assumed spread capture; the directional component priced at the mid is −Rs 10.14 lakh. The win rate falls from 77% on the simulator to 12.6% at the mid. The strategy has no directional edge; it is a pure spread harvester whose “wins” are the spread assumption.

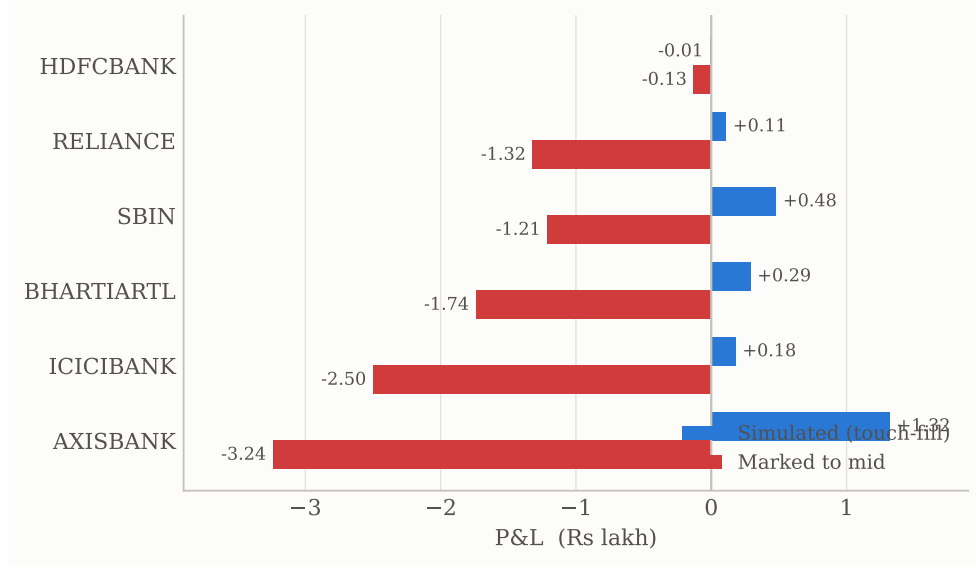


Figure 2. Per-instrument profit, simulated versus marked to mid. The spread-capture illusion is present in every name; the strongest apparent earners are the largest at-mid losses.

The pattern is universal across the universe (Figure 2), not driven by one name. The apparent top earner, AXISBANK at +Rs 1.32 lakh, is −Rs 3.24 lakh once priced at the mid; every instrument flips negative.

The markout curve (Figure 3) sharpens the diagnosis. The entry markout is positive and roughly flat out to sixty seconds (Rs 197 at the fill, Rs 176 at 60 s), so the favourable fill is real at the instant of execution and the entry is *not* catastrophically adversely selected at these horizons. The loss does not accrue at entry; it accrues between entry and exit, where the mid drifts against the position and only the spread assumption keeps the trade in the black.

Finally, the decomposition sets a hard target (Figure 4). Writing realistic net as $P\&L_{\text{mid}} + f \cdot (\text{spread capture})$ for a retained spread fraction f , break-even occurs at $f^* = 81\%$. Independently, a queue-aware fill simulation placed the break-even maker-exit fill rate near 67 to 80%, corroborating the threshold by a second, unrelated method.

6 Statistical rigour: subtracting luck

A multi-arm race makes the best arm look good by chance alone, so no arm was trusted on its raw numbers. Arms were ranked on cost-realistic, out-of-sample returns using the effective sample size n_{eff} (trades are autocorrelated, so the raw count overstates information), the Probabilistic and Deflated Sharpe Ratios [2, 3], and the Probability of Backtest Overfitting via combinatorially-symmetric cross-validation [4]. The promotion gate was $\text{DSR} \geq 0.95$ and $\text{PBO} \leq 0.2$.

The best arm reached a Deflated Sharpe of 0.93, short of the gate, on returns that were themselves optimistic; the Probability of Backtest Overfitting was near zero, indicating the *selection* was sound while the *edge* was not. The statistical verdict and the execution-realism finding agree: there was nothing to promote.

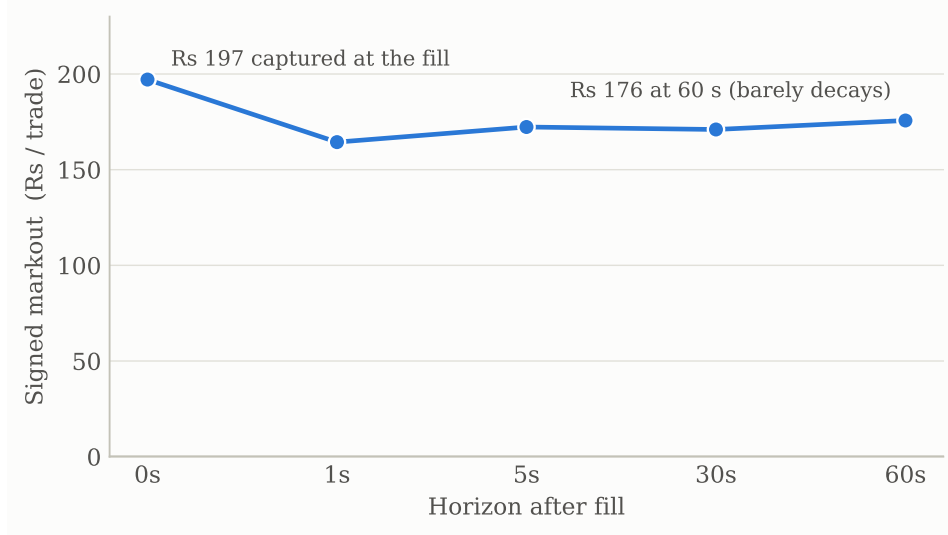


Figure 3. Signed entry markout by horizon. The half-spread captured at the fill is real and barely decays, so entry toxicity is not the mechanism; the negative result lives in the exit-side price drift that the mid-to-mid view exposes.

7 Interpretation: a structural, seat-specific verdict

The strategy is a pure spread harvester with a negative directional edge, so its survival depends entirely on spread retention, which is governed by queue position. A retail seat cannot reach the required 81%. It sits at the back of a first-in-first-out queue without colocation, acts at roughly 15 ms latency, earns no maker rebate on this segment, and pays a securities transaction tax (STT) that is about two-thirds of round-trip cost and falls on a seconds-hold, one-tick-spread strategy that cannot out-earn it (Table 1).

Table 1. Per-instrument cost, one lot. STT scales with the sell-side notional, so the break-even move per share, not a flat threshold, is the economically meaningful quantity, and it varies roughly threefold across the universe.

Instrument	Round-trip fee	Break-even move / share
HDFCBANK	~ Rs 126	Rs 0.23
RELIANCE	~ Rs 177	Rs 0.35
ICICIBANK	~ Rs 222	Rs 0.32

The subtle point is *where* the adverse selection lives. It is not mainly in the markout horizon, which is flat-positive; it is in fill *selection*. From the back of the queue you miss the benign touches (price taps your level and bounces, the resting orders ahead of you absorb it, and you do not fill) and you keep the adverse breaks (price trades through, the queue clears, and you fill an instant before the move continues against you). The touch-fill model deletes this bias by construction because it fills on every touch; pricing at the mid restores it. Relatedly, the microprice does predict the next tick, but the colocated participants who set it act microseconds after the book changes, so at retail latency one trades *after* the information is already in the price. The signal is real; the seat cannot reach it.

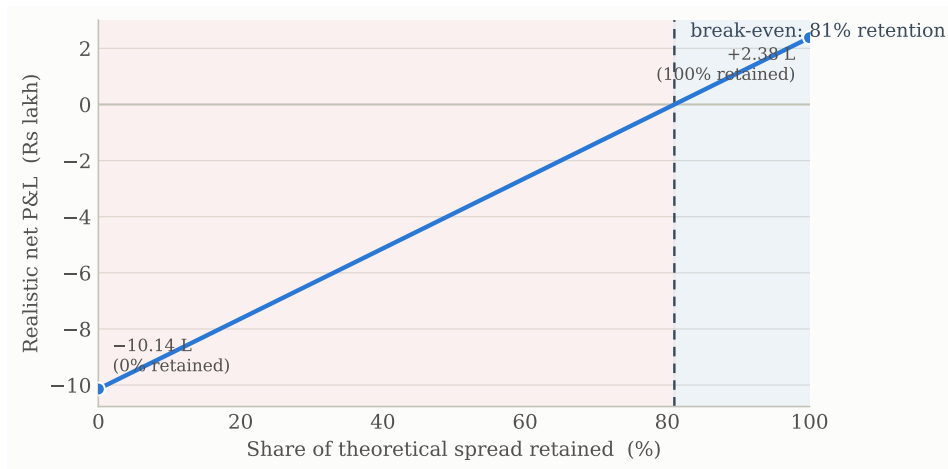


Figure 4. Profitability as a function of how much of the theoretical spread is actually retained. Break-even needs 81% retention; realistic execution must land far toward the favourable end of this line, which is the hard direction to reach, not the easy one.

8 The pivot: seat-first strategy design

Rather than tune a structurally dead strategy, the design was re-derived from the seat outward: given no colocation, no exchange membership, no rebates, roughly 15 ms latency, and small capital, which edges are even available? The market-making (latency-race) game is closed to that seat. What remains open is *taker-side*, *model-based* strategies at horizons where latency does not decide the outcome: statistical arbitrage and Ornstein-Uhlenbeck mean-reversion, principal-component residual reversion [6], cross-sectional short-horizon reversal [8], and calendar spreads, each derived from a testable statistical property (cointegration [7], factor structure, order-flow impact [5]) rather than a tuned chart indicator. A three-year daily and one-minute historical dataset was assembled as the foundation for that program.

9 Conclusion

The market-making strategy has no viable edge for a self-funded retail seat, and its only path to profitability requires institutional execution infrastructure, colocation, FPGA order handling, and exchange membership, that is outside a bootstrapped budget. The identified pivot is a longer research program that would need either risk capital or more runway. Given that, the disciplined choice was to halt at a clean, documented stopping point rather than deploy capital into an unvalidated strategy.

The value delivered is the research itself: a method (the markout and mid-to-mid decomposition, paired with overfitting-aware evaluation) that separates a real edge from a fill-model artifact; a finding (a strategy that looked profitable was falsified) reached on paper with zero capital at risk; and the judgment to stop. A false edge is cheapest to discover before it is funded.

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All trading described in this report was simulated on live and historical market data. No orders were sent to a live exchange and no real capital was deployed.